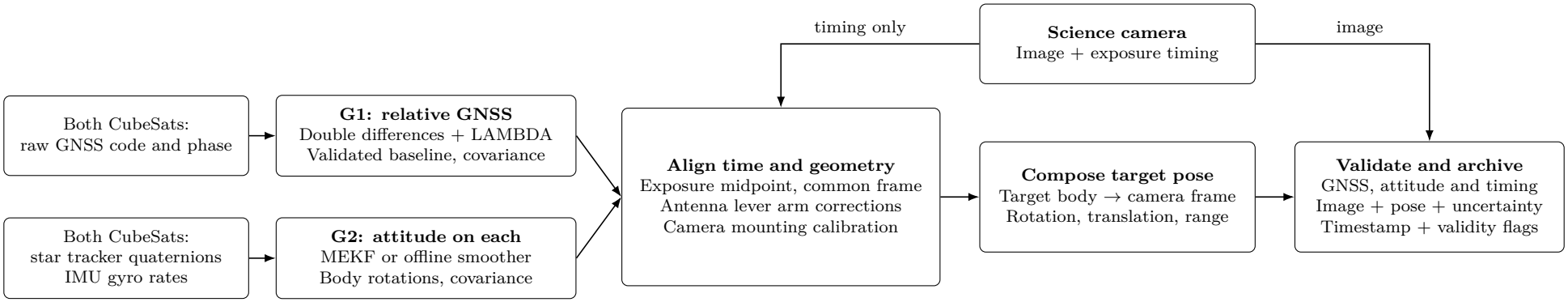
CubeSat image dataset: camera and independent pose reference alternatives

Assumptions: two cooperative spacecraft in low Earth orbit, with a star tracker and an IMU on each. Reference pose uses GNSS and attitude telemetry. No visual targets or docking features are added. Linked option names lead to sources.

Alternative	Capability or measurement method	Mission implication
Image cameras (each meets the 1920×1200 minimum)		
C1. GomSpace NanoCam C1U	2048×1536 RGB CMOS. Demonstrated in orbit on Delphini-1.	Heritage reference. The manufacturer lists it as phased out. Verify exposure timing and shutter mode, and focus the optics for proximity.
C2. MSSS ECAM-C50	2592×1944 RGB, rolling shutter. Native format: 2.8 frames/s over 200 Mbit/s SpaceWire. Imaging power: 1.6 W.	Flight proven on OSIRIS-REx. Rolling shutter needs motion assessment. Optics and baffle can dominate the CubeSat volume allocation.
C3. FLIR Blackfly S BFS-U3-51S5C-C	2448×2048 RGB, global shutter. Up to 73 frames/s in 8-bit Bayer format. 36 g without lens, 3 W maximum.	Compact industrial camera with hardware I/O for synchronization. Requires mission qualification for radiation, vacuum and temperature.
Independent reference sensors and algorithms (exclude the training images from label generation)		
G1. Raw carrier-phase GNSS on both spacecraft	Relative translation and range. Double-differenced carrier phase → LAMBDA integer ambiguity resolution → weighted position solution and optional offline smoothing.	Pair with G2 for full pose. Requires common GNSS visibility and validated ambiguity fixes. A published CubeSat hardware ground test achieved about 1.5 cm position RMSE. This is not an orbital accuracy guarantee.
G2. Existing star trackers and IMUs	Body attitude in a common inertial frame. Star tracker quaternions and IMU gyro rates → multiplicative extended Kalman filter (MEKF) or offline attitude smoother → relative rotation composition.	Uses existing sensors. Estimate gyro bias and calibrate sensor and camera alignments. Relative pose uncertainty includes both attitude solutions, synchronization error and mounting drift. Validate tracker coverage over the required slew rates.

Pose label generation with G1 + G2



Frame composition. Let O denote the observer body, T the target body, C the camera and I a common inertial frame. The transform ${}^{X}T_{Y}$ maps points from Y to X. G1 supplies the differential translation, G2 supplies the body rotations, and calibrated antenna offsets locate the body origins. Use the calibrated camera transform ${}^{O}T_{C}$ for the active optical setting.

$$ {}^{C}T_{T} = \left({}^{I}T_{O}\, {}^{O}T_{C} \right)^{-1} {}^{I}T_{T}, \qquad \text{distance} = \left\| {}^{C}\mathbf{t}_{T} \right\|. $$

Operation and validation. Process synchronized telemetry offline, or exchange it over an intersatellite link for onboard processing. Image pixels enter only the archive. Calibrate camera intrinsics and sensor geometry. Publish timestamps, covariance and validity flags.

Literature reference. SPEED+ (Sec. 5, p. 8, Eq. 4) uses calibration-based scoring thresholds of $\theta_t = 2.173\,\text{mm/m}$ for normalized three-dimensional translation and $\theta_q = 0.169°$ for rotation. Adopt these as provisional total label error ceilings: $B_t(r) = 2.173 \times 10^{-3} r$ and $B_q = 0.169°$, where r is camera-to-target range. Lateral error shares the total translation budget. These thresholds specify neither a confidence level nor a one-pixel criterion.

Close range implication. CubeSat-CDT (Table 2, p. 116) reports real-image ranges of 0.65–1.2 m, yielding $B_t = 1.41$–2.61 mm under the adopted criterion. Its Table 4 (p. 122) reports model prediction errors. The paper does not quantify label uncertainty. The cited 15 mm GNSS RMSE does not demonstrate this accuracy. Allocate and validate contributions from GNSS, attitude, timing and calibration with stated confidence and margin.

Complement: optical field of view and 2U framing distance

Geometry (nominal pinhole model). Use the bare CDS Rev. 14.1, Appendix B, 2U drawing envelope, $W \times D \times L = 0.100 \times 0.100 \times 0.227$ m, including the rail ends. The target body center lies on the camera boresight. Distances below are from the camera optical center to the target center. All calculations use the full native sensor, with the shorter field of view $\theta_s = \theta_V$.

$$\theta_H = 2\arctan\left(\frac{N_H p}{2f}\right), \qquad \theta_V = 2\arctan\left(\frac{N_V p}{2f}\right), \qquad d_{\text{fill}} = \frac{D}{2} + \frac{L}{2\tan(\theta_s/2)}, \qquad d_{\text{all}} = \frac{\sqrt{W^2 + D^2 + L^2}}{2\sin(\theta_s/2)}.$$

Meaning. d_{fill} is the minimum framing distance with the 227 mm edge aligned with the shorter frame dimension and the 100 mm depth along the boresight. At equality, the near face fills the shorter dimension. d_{all} contains the target's circumscribed sphere and guarantees geometric fit for every attitude. Larger distances also fit. An upper range requires a minimum target size in pixels.

Optics. NanoCam and ECAM entries are discrete fixed lens configurations. Blackfly S requires a separate C-mount lens. The optional Computar lens below provides a continuous 3× motorized optical varifocal range, with motorized refocusing.

Camera and lens	f [mm]	H FOV [°]	V FOV [°]	d_{fill} [m]	d_{all} [m]
C1. NanoCam C1U + Schneider Cinegon	8.2	43.56	33.37	0.429	0.466
C1. NanoCam C1U + Schneider Xenoplan	34.9	10.73	8.06	1.662	1.904
C1. NanoCam C1U + Schneider Tele-Xenar	70.5	5.32	3.99	3.306	3.838
C2. ECAM-C50 + factory fixed optic	3.85	80	62	0.239	0.260
C2. ECAM-C50 + factory fixed optic	7.1	44	33	0.433	0.471
C2. ECAM-C50 + factory fixed optic	12.6	25	19	0.728	0.810
C2. ECAM-C50 + factory fixed optic	29.7	11	8	1.673	1.917
C2. ECAM-C50 + factory fixed optic	63	5	4	3.300	3.832
C3. Blackfly S + Kowa LM12JC10M (fixed)	12	38.77	32.81	0.436	0.474
C3. Blackfly S + Computar ML3Z1226UC-MPW (zoom)	12 → 36	38.77 → 13.38	32.81 → 11.21	0.436 → 1.207	0.474 → 1.369

FOV basis. C1 angles are calculated using the MT9T031 sensor's 3.2 μm pixels and 2048×1536 active array. C2 angles are reported directly by MSSS and rounded by the supplier. C3 angles are calculated using the Blackfly S 3.45 μm pixels and 2448×2048 active array. The Computar specification's example FOV uses a different 8.8×6.6 mm sensor, so its angles are recalculated for Blackfly S. Arrows show wide-to-tele settings.

Focus and integration. Kowa specifies a 0.10 m minimum working distance. Computar specifies focus ranges from 0.10 m at wide and 0.35 m at tele, with a 247 g lens and USB motor control. These lenses require mission qualification. Confirm focus for C1 and C2, and calibrate usable FOV, distortion, focus breathing and camera extrinsics at each optical setting. Add pointing, framing and collision margins. Include appendages and protrusions in the final CAD envelope.

Finite upper range. For the aligned case, a chosen minimum target span $0 < n_{\min} \le N_V$ pixels on the shorter sensor side gives $d_{\max} = D/2 + \dfrac{N_V L}{2n_{\min}\tan(\theta_s/2)}$. Without $n_{\min}$, FOV supplies a minimum fit distance only. For the 12–36 mm optical zoom, the farthest aligned exact-fill distance is approximately 1.21 m. At the all-attitude limits of 0.474–1.369 m, the SPEED+ planning translation ceiling from page 1 is 1.03–2.98 mm.